

HOW PGL FINDS A SWEET SPOT IN PHOSPHOLIPID MEMBRANES - A COMBINED MULTISCALE MD AND NMR STUDY

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ABSTRACT

Glycolipids from pathogenic *Mycobacterium tuberculosis* (*Mtb*) play important roles during the interaction of the pathogen with macrophages, and can shape the host cell's immune response by modulating its membrane structure and function. Here we study the phenolic glycolipids (PGLs) present in the envelope of some hypervirulent strains of *Mtb*, and their impact on model membranes. By a combination of molecular modeling and simulations, and solid-state NMR experiments, we show that PGLs, like the structurally related lipid PDIM, adopt a conical shape in lipid membranes which destabilizes the lamellar membrane phase and promotes a transition to a non-lamellar inverted-hexagonal phase. Unlike PDIM, in our simulations PGLs remain anchored to the phosphate groups of the lipid bilayer by its sugar-carrying extremity, preventing lipid flip-flop. These findings shed new light on a potential biophysical role of PGLs, through modulation of the properties of the host cell's membrane, in addition to the recognition of its sugar moiety by host cell immune receptors.

REFERENCES

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